④ Ex - ~~spam det~~ stock price prediction, house price prediction, demand forecasting	④ Ex - spam detection, image recognition, sentiment analysis
⑤ Algo - linear regression, polynomial regression	⑥ Algo - Logistic Regression, Decision vector.

3→ Decision tree is a flowchart-like structure used to make decisions or predictions. It consists of nodes representing decision or tests on attributes, branches representing the outcome of these decisions & leaf nodes representing final outcome or predictions.

08 12⁴ Unit 2

- Logistic Regression
- Logistic regression algo is widely use for binary classification problem. It models the probability that a given input belong to a particular class
- Unlike to linear regression which predicts a continuous outcome, logistic regression predict a binary outcome (0 or 1) by modeling the relationship between the input features & the probability of particular outcome using logistic functions
- Logistic regression is one of the most popular supervised learning Algo. It is used for predicting the categorical dependent variable using the given set of independent variable
- Logistic regression predict the output of categorical dependent variable & hence the outcome must be categorical discreate value, such as yes or no, 0 or 1, true or false.
- It gives the probabilistic value which lies between 0 & 1
- In Logistic regression "S-shape" can be seen after fitting the values in visualition
- The curve from logistic regression indicates the likelyhood of some values

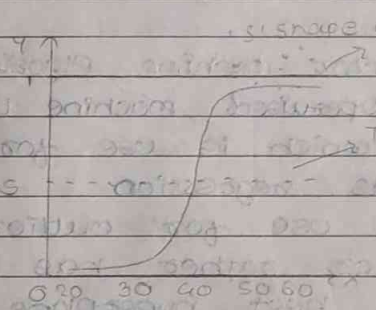

Fig: Logistic regression

- Sigmoid function
- Logistic regression uses sigmoid funⁿ which is mathamatical funⁿ use to map predicted values to probability

$$y = \frac{1}{1+e^{-(mx+b)}}$$

- The value of logistic regression must be between 0 & 1 which cannot go beyond this limit.
- It forms S-curve shape which is called sigmoid function or logistic function.
- In Logistic regression there are some certain assumtions such as the dependent variables must be categorical in nature & independent variable should not be multicolinear.
- Logistic regression has three types
  1. Binomial
  2. Multinomial
  3. Ordinal

06/08/24

- Support vector machine Algorithm
- SVM is supervised machine Learning Algorithm which is use for classification as well as regression
- It can be use for outliers detections
- It works or rather has goal to find the best hyperplane. It which separate datapoints of different classes in a high dimensional space
- The hyperplane in SVM is also called as decision boundry that can segregate dimensional space into classes in such a way the new datapoint can be easily categorised & predict that class it belongs.
- SVM selects the datapoints (extream datapoint) to create hyperplane.
- This datapoints are called as support vectors

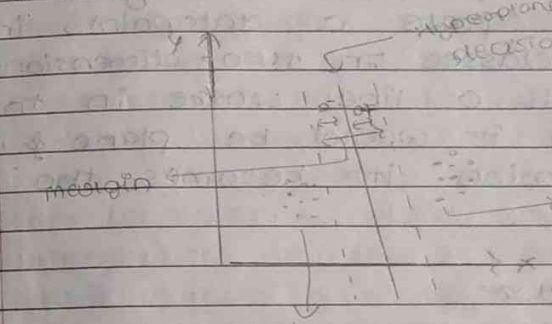

Fig: SVM

- Types of SVM
  Generally there are two types of SVM
  1. Linear SVM
  2. Non-linear SVM

1. Linear SVM
   It is use for linearly separable data i.e dataset which can be classify into two classes with single straight line & linear SVM classifier is used.

2. Non-linear SVM
   It is use for non-linearly separable da

i.e dataset cannot be classify by using straight line

- Terminologies of SVM

① Hyperplane or decision boundary
In SVM a decision boundary or hyperplane seprate a datapoints in different classes. In two-dimensions space it is a line where in three dimensions it would be plane & in higher dimension it becomes the hyperplane

② Support vectors
Support vectors are datapoints which are close to hyperplane.
Hyperplane is selected to maximise the margin between support vectors of different classes

③ Margin
It is the disⁿ (total distence) betn the hyperplane & support vectors from their classes.

SVM has aim to maximise the margin to ensure that model is robust & can generalise well to new data or unseen data

- Kernal function & Kernal SVM

- In SVM the Kernal function is used to transform the input data into a higher dimensional space.
- This transformation make it easier to find linear seperation which is nothing but hyperplane or decision boundry between the classes in the new space even if the data is not linearly seperable in the original space. The kernal funⁿ perform this transformation without much computing coordinates in the higher dimensional space which is called kernal trick.

- Types of kernal function in SVM

① Linear Kernal
This type of kernal funⁿ used is ML including SVM. It is the simplest kernal funⁿ which is suitable for linearly seperable data.
$K(x, y) = x \cdot y$
where x & y are input feature vector & dot product of input vector measures the similarity or the distance in the original feature space

② polynomial kernal
It represents the similarity of vectors in a feature space over polynominal of original variables.

$K(x, y) = (xy + c)^d$

where c is a constant that adjust the influence of higher terms & d is degree of polynomial.

This polynomial kernal is use when the interaction between ~~the~~ features are polynomial

Kernals are fundamental to the power of SVM allowing them to classify data that is non-linearly separable while transforming into a higher dimensional space.

Selection of kernals & its parameter can significantly impact the performance of SVM model.